# Sai Sravan Vadapalli

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## EDUCATION

| Institution | Degree / Board | Passing Year | Score / CGPA |
|---|---|---|---|
| Indian Institute of Technology Hyderabad | B.Tech | 2027 | 8.52 |
| Gyanavapi Junior College | Class XII (Telangana State Board of Intermediate Education) | 2017 | 94.90% |
| Kenedy Vidya Bhavan School | Class X (Board Of Secondary Education Telangana State) | 2015 | 9.00% |

## EXPERIENCE

**IIT Hyderabad | Practical Networking and Blockchain (PRANET) Lab | Research Assistant**

Supervisor: Prof. Kotaro Kataoka

- Implementing a **multi-agent LLM** system for **automated network fault diagnosis**, coordinating specialized agents over a **Neo4j Knowledge Graph (KG)** backed trouble ticket repository.
- Validating the system on emulated SDN topologies under **injected faults, grounding diagnoses** in live **Digital Twin state.**

- Implemented a data-representation based **Digital Twin Network (DTN)** architecture for **SDN**, using a **Neo4j Knowledge Graph (KG)** for network state storage and cutting build time by 15% via MERGE-based Cypher optimization.
- Built a **Network State Verification (NSV)** framework extending the KG-based DTN to **validate new configurations** with zero risk to the live network.

**Accenture | Custom Software Engineering Associate**

- Developed **Spring Boot microservices** for a European pay-TV broadcaster's website, with **Kafka** handling event-driven messaging and **Couchbase** providing low-latency data access.
- Migrated production microservices from **Akka** to **Spring Boot**, reducing complexity and improving maintainability.
- Built **REST APIs** across the microservices and set up a **local sandbox with Microcks**, mocking cross-team dependencies to unblock independent development and testing.
- Debugged and resolved production issues using **GCP Cloud Logging** for centralized log analysis across services.

## PROJECTS

**LLM Inference Routing: Survey & Benchmarking| IITH - Marvell AI/ML Networking & Infrastructure Workshop 2025 | Github**

- Benchmarked **GraphRouter**, a **GNN-based LLM router**, measuring **~57 µs inference per query** and validating reward gains over **FrugalGPT**, **HybridLLM**, and **C2MAB-V** baselines.
- Surveyed **inference-time LLM routing strategies** (pre-generation, post-generation, and cascade) across cost, latency, and accuracy trade-offs.

**Fraud Detection Using Variational AutoEncoders |** Python, TensorFlow, Scikit-learn, Pandas, Matplotlib **| Github**

- Built an **unsupervised fraud detection** system using **Autoencoders** and **Variational Autoencoders** to flag anomalous credit card transactions from reconstruction error, trained only on legitimate transaction data.
- Benchmarked 4 model architectures (shallow/deep AE and VAE) on 284K+ real-world transactions, tuning error thresholds via mean + k·std to achieve up to **99.3% precision** and **0.90 F1-score** in fraud identification.

**Fraud Detection using Trust-Based Ranking |** Python, Pandas, NetworkX, Scikit-learn **| Github**

- Built a fraud-detection system for payment networks by implementing the **TrustRank algorithm** to score account trustworthiness from transaction graphs.
- Modeled ~800 accounts as a **payment graph**, seeded trust via **inverse PageRank** validated against a fraud oracle, and propagated scores that correctly ranked all 20 known fraudulent accounts lowest.

**Secure Peer-to-Peer Chat Application |** C++, OpenSSL, DTLS 1.2, Docker **| Github**

- Built a peer-to-peer encrypted chat application in C++ using **DTLS 1.2** over **UDP** with mutual **X.509 certificate authentication**.
- Analyzed how a single compromised Certificate Authority undermines **PKI-based mutual DTLS trust**, by engineering a **downgrade attack** and an **active MITM proxy** using forged shadow certificates that decrypted and tampered with **secure traffic** undetected.
- Validated both attacks under simulated packet loss across containerized network topologies using **Docker**.

## PUBLICATIONS

Dnyaneshwar Shinde, **Sai Sravan V**, Kotaro K. "Integrating Network State Verification with Knowledge Graph based Digital Twin for SDN," published in 11th IEEE Conference on NFV-SDN 2025, Athens, Greece, November 2025.

## TECHNICAL SKILLS

**Languages**: Python, C++, C, Core Java, SQL, Bash Scripting, Neo4j Cypher

**Frameworks & ML Libraries**: Spring Boot,Apache Kafka, Langgraph, RAG, RYU, Mininet, Scikit-learn, NumPy, Pandas, Matplotlib, Seaborn

**Networking Tools**: Wireshark, tcpdump, Cisco Router, Vyos Router, Zeek, Burpsuite

**Databases**: Neo4j, MySQL, CouchBase,PostgresSQL

**Cloud & DevOps**: Docker, Kubernetes, Git, Github

## RELEVANT COURSES

**Theory & Concept**: Advanced Data Structures and Algorithms,  Object Oriented Programming

**Systems & Networking**: Advanced Computer Networks, Network Engineering, Network Security

**AI &ML:** Foundations of Machine Learning, Deep Learning, Fraud Analytics, Multimedia Content Analysis

## SCHOLASTIC ACHIEVEMENTS

- Selected as a **Research Exchange Student** at the **University of Tokyo, Japan** (Dec, 2024)
- Secured **Elite + Silver Certification(84%)** in *Programming in Java*, a 12-week course conducted by **IIT Kharagpur** via NPTEL.
- Secured **Elite + Silver Certification(82%)** in *Deep Learning*, a 12-week course conducted by **IIT Madras** via NPTEL.

## INTERESTS & HOBBIES

Football, Gaming, Reading novels